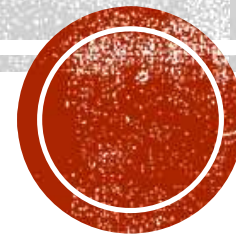


WINNING THE SPACE RACE WITH DATA SCIENCE



OUTLINE

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EXECUTIVE SUMMARY

- This project focuses on predicting the successful landing of the Falcon 9 first-stage booster using data science techniques. SpaceX launch data was collected from the SpaceX API and supplemented with historical records obtained through web scraping from Wikipedia. The data was cleaned, processed, and analyzed using exploratory data analysis and SQL queries. Interactive visualizations were created using Folium and Plotly Dash to explore launch sites, payload mass, and success rates. Several machine learning models, including Support Vector Machine, K-Nearest Neighbors, and Decision Tree, were developed and evaluated. Among these models, the Decision Tree achieved the highest prediction accuracy, making it the most effective model for this task.



INTRODUCTION

- SpaceX has transformed the aerospace industry by significantly reducing launch costs through the reuse of the Falcon 9 first-stage booster. Accurately predicting whether the first stage will successfully land is critical for estimating launch costs and supporting strategic decision-making for companies competing in the commercial space market.
- This project aims to analyze historical SpaceX launch data to identify the key factors that influence first-stage landing success. By applying data science techniques and machine learning models, the project seeks to build an effective predictive system and determine which model performs best in predicting Falcon 9 landing outcomes.



DATA COLLECTION AND DATA WRANGLING METHODOLOGY

- SpaceX launch data was collected using the official SpaceX REST API and complemented with historical launch records obtained through web scraping from Wikipedia. These two data sources were integrated to create a comprehensive dataset covering launch details, payload information, launch sites, and landing outcomes.
- Data wrangling involved cleaning and preprocessing the dataset by handling missing values, correcting data types, and standardizing formats for consistency. Additional features were engineered to support analysis, and the final dataset was validated to ensure accuracy and reliability before being used for exploratory analysis and machine learning modeling.



EDA AND INTERACTIVE VISUAL ANALYTICS METHODOLOGY

- Exploratory Data Analysis (EDA) was performed to understand the distribution of variables and identify patterns related to Falcon 9 first-stage landing success. Visualizations such as bar charts, line plots, scatter plots, and heatmaps were used to examine relationships between launch outcomes, payload mass, launch sites, orbit types, and flight history. SQL queries were also applied to summarize launch statistics and success rates.
- Interactive visual analytics were developed to enhance data exploration. Folium was used to create interactive maps showing launch site locations and landing success rates, allowing geographic patterns to be analyzed. In addition, an interactive dashboard was built using Plotly Dash, enabling users to dynamically explore launch success in relation to payload mass and launch sites through dropdown menus and sliders.



PREDICTIVE ANALYSIS METHODOLOGY

- Predictive analysis was conducted by first preprocessing the data, including feature scaling and splitting the dataset into training and test sets. Several classification models were then implemented, including Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Decision Tree, to predict the success of Falcon 9 first-stage landings.
- Hyperparameters for each model were optimized using GridSearchCV to improve performance. The models were evaluated using test data accuracy, and their results were compared to identify the most effective approach for predicting landing success.



EDA WITH VISUALIZATION RESULTS

Exploratory Data Analysis with visualizations revealed several important insights into Falcon 9 first-stage landing success. Analysis across different **launch sites** showed varying success rates, indicating that launch location plays a significant role in landing outcomes. Visualizations also demonstrated that **landing success improves over time**, suggesting that operational experience and technological improvements contribute to higher reliability.

Payload-related visualizations showed that **successful landings are more frequent for lower payload masses**, while heavier payloads exhibit a mix of successes and failures, indicating increased mission complexity. Orbit-based analysis revealed that most orbit types achieved high success rates, whereas **GTO missions showed comparatively lower success**. Overall, these visual results highlight clear relationships between launch site, payload mass, orbit type, and landing success, providing a strong foundation for predictive modeling.



EDA WITH SQL RESULTS

Exploratory Data Analysis using SQL was conducted to extract structured insights from the SpaceX launch dataset. SQL queries were used to calculate the **total number of launches**, as well as the **number of successful and unsuccessful missions**, providing an overall view of launch performance. Additional queries computed **success rates by launch site**, allowing comparison of performance across different locations.

Further SQL analysis focused on **payload-related metrics**, including total and average payload mass by rocket and booster version. Time-based queries were used to examine launch records within specific periods and to identify trends in mission outcomes over time. Ranking and filtering queries were applied to compare landing outcomes across different date ranges and mission configurations.

Overall, the SQL-based EDA results complemented the visual analysis by providing precise, query-driven evidence of trends and patterns in launch success. These results confirmed that launch site characteristics, booster versions, and mission profiles significantly influence Falcon 9 first-stage landing outcomes.



INTERACTIVE MAP WITH FOLIUM

Interactive maps were created using **Folium** to visualize SpaceX launch site locations and landing outcomes. All launch sites were marked on the map with location-based markers, allowing clear geographic identification of where launches occurred.

Launch outcomes were visually distinguished using color-coded markers to represent **successful and unsuccessful landings**, enabling quick comparison of success rates across different sites. In addition, distance calculations and proximity features were included to show the relationship between launch sites and nearby coastlines or facilities. These interactive maps support deeper geographic analysis and provide clear insights into spatial patterns related to Falcon 9 launch success.



INTERACTIVE DASHBOARD WITH PLOTLY DASH

An interactive dashboard was developed using **Plotly Dash** to analyze Falcon 9 launch performance. The dashboard includes a **pie chart** displaying the overall distribution of successful and unsuccessful landings and a **scatter plot** illustrating the relationship between payload mass and landing success.

Interactive components such as a **launch site dropdown menu** and a **payload mass range slider** were implemented, allowing users to dynamically filter and explore the data. These interactive features enable deeper analysis of how launch site selection and payload characteristics influence Falcon 9 first-stage landing success.



PREDICTIVE ANALYSIS RESULTS

The predictive analysis results demonstrate the effectiveness of machine learning models in forecasting Falcon 9 first-stage landing success. Multiple classification models were evaluated, including **Support Vector Machine (SVM)**, **K-Nearest Neighbors (KNN)**, and **Decision Tree**. Model performance was assessed using test data accuracy as the primary evaluation metric. The **Decision Tree model achieved the highest accuracy (94.44%)**, outperforming SVM and KNN, which both achieved an accuracy of **83.33%**. The results indicate that the Decision Tree model is the most reliable and interpretable approach for predicting landing outcomes. These findings confirm that machine learning can successfully capture key factors influencing Falcon 9 landing success and provide strong predictive capability.



CONCLUSION

- This project demonstrated how data science and machine learning techniques can be effectively applied to predict the successful landing of the Falcon 9 first-stage booster. Through data collection, exploratory analysis, interactive visualizations, and predictive modeling, key factors influencing landing success were identified.
- The results showed that launch site characteristics, mission profiles, and operational experience play an important role in landing outcomes. Among the evaluated models, the Decision Tree achieved the highest accuracy, highlighting its effectiveness for this task. Overall, the insights and tools developed in this project provide a strong foundation for future analysis and decision-making in reusable rocket operations.



THANK YOU!

